FURTHER MECHANICS / KINEMATICS & DYNAMICS

Motion Under a Variable Force

Worked solutions 14 questions · 162 marks

EXAM BOARDS Edexcel FM2 9FM0/4C · OCR A Y543 · OCR MEI Mech Major Y421
CIE 9231 Paper 3

Instructions

Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question gives a value. Where a question leaves g unspecified and a final answer depends on it, the solutions also give that answer for $g = 10\,\mathrm{m\,s^{-2}}$.

CIE (9231 Paper 3): papers take $g = 10\,\mathrm{m\,s^{-2}}$; use the $g = 10$ final answers where the solutions give them.

Questions

Question 1	*9 marks*	2
Question 2	*7 marks*	4
Question 3	*9 marks*	5
Question 4	*13 marks*	7
Question 5	*14 marks*	9
Question 6	*16 marks*	11
Question 7	*9 marks*	13
Question 8	*11 marks*	15
Question 9	*12 marks*	17
Question 10	*10 marks*	19
Question 11	*13 marks*	21
Question 12	*9 marks*	23
Question 13	*14 marks*	25
Question 14	*16 marks*	28

1. A particle P of mass $4\,\text{kg}$ moves along a straight horizontal line. At time $t\,\text{s}$, its displacement in the positive direction from a fixed point O is $x\,\text{m}$ and its velocity is $v\,\text{m}\,\text{s}^{-1}$.

The only horizontal force acting on P has magnitude

$$\frac{36}{(x+2)^3}\,\text{N}$$

and acts in the direction of decreasing x. Initially, P is at O and is moving in the positive direction with speed $\frac{3}{2}\,\text{m}\,\text{s}^{-1}$.

(a) Find an expression for v in terms of x. **[5]**

(b) Hence find an expression for x in terms of t. **[4]**

Solution

(a) Take the direction of increasing x as positive. Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$4v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{36}{(x+2)^3}$$

Separating the variables and integrating:

$$\int 4v\,\mathrm{d}v = -36\int (x+2)^{-3}\,\mathrm{d}x$$

$$2v^2 = \frac{18}{(x+2)^2} + c$$

Initially, $x = 0$ and $v = \frac{3}{2}$, so

$$2\left(\frac{3}{2}\right)^2 = \frac{18}{2^2} + c \quad \Longrightarrow \quad c = 0$$

Thus $v^2 = \frac{9}{(x+2)^2}$. The velocity is initially positive and cannot become zero at any finite $x \geq 0$, so we take the positive root:

$$v = \frac{3}{x+2}$$

(b) Using part (a) and $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, separate the variables and integrate:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3}{x+2}$$

$$\int (x+2)\,\mathrm{d}x = \int 3\,\mathrm{d}t$$

$$\frac{x^2}{2} + 2x = 3t + d$$

Since $x = 0$ when $t = 0$, $d = 0$. Completing the square gives

$$x^2 + 4x = 6t$$

$$(x+2)^2 = 6t + 4$$

Taking the positive root since $x + 2 > 0$:

$$x = \sqrt{6t + 4} - 2, \qquad t \geq 0$$

2. A lorry of mass 2000 kg travels along a straight horizontal road. Its engine delivers constant power of 80 kW. When the lorry is moving at speed $v\,\mathrm{m\,s^{-1}}$, the resistance to its motion has magnitude $200v$ N.

At time t seconds, the speed of the lorry is $v\,\mathrm{m\,s^{-1}}$. At $t = 0$, the speed of the lorry is $8\,\mathrm{m\,s^{-1}}$.

(a) Show that

$$10v\frac{\mathrm{d}v}{\mathrm{d}t} = 400 - v^2 \qquad \textbf{[3]}$$

(b) Verify that, while $8 \leq v < 20$,

$$t = 5\ln\left(\frac{336}{400 - v^2}\right) \qquad \textbf{[4]}$$

Solution

(a) Take the direction of motion as positive. Using $P = Fv$, the driving force is

$$F = \frac{80\,000}{v}\,\mathrm{N}$$

Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$2000\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{80\,000}{v} - 200v$$

Multiplying by $\frac{v}{200}$:

$$10v\frac{\mathrm{d}v}{\mathrm{d}t} = 400 - v^2$$

(b) Separating the variables in the equation from part (a):

$$\frac{10v}{400 - v^2}\,\mathrm{d}v = \mathrm{d}t$$

For $8 \leq v < 20$, $400 - v^2 > 0$. Since the derivative of $400 - v^2$ is $-2v$, integrating gives

$$\int 1\,\mathrm{d}t = -5\int \frac{-2v}{400 - v^2}\,\mathrm{d}v$$

$$t = -5\ln(400 - v^2) + c$$

Using $v = 8$ when $t = 0$:

$$0 = -5\ln(400 - 8^2) + c$$

$$c = 5\ln 336$$

Substituting and combining the logarithms:

$$t = 5\ln 336 - 5\ln(400 - v^2)$$

$$= 5\ln\left(\frac{336}{400 - v^2}\right)$$

3. A particle P of mass $0.5\,\text{kg}$ moves along the positive x-axis under the action of a single force. At time t seconds, the particle is x metres from the origin O, where $x > 0$, and has speed $v\,\text{m s}^{-1}$. The force acts in the direction of increasing x and has magnitude

$$\frac{k}{x^3} \text{ newtons}$$

where k is a positive constant.

When $x = 2$, the speed of P is $1\,\text{m s}^{-1}$, and when $x = 4$, its speed is $2\,\text{m s}^{-1}$.

(a) Show that

$$v^2 = 5 - \frac{16}{x^2} \qquad \textbf{[5]}$$

The time taken for the speed of P to increase from $1\,\text{m s}^{-1}$ to $2\,\text{m s}^{-1}$ is T seconds.

(b) Use algebraic integration to show that

$$T = \frac{6}{5} \qquad \textbf{[4]}$$

Solution

(a) Take the direction of increasing x as positive. Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$\frac{1}{2}v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{k}{x^3}$$

Separating the variables and integrating:

$$2v\,\mathrm{d}v = 4kx^{-3}\,\mathrm{d}x$$

$$\int 2v\,\mathrm{d}v = \int 4kx^{-3}\,\mathrm{d}x$$

$$v^2 = -\frac{2k}{x^2} + c$$

Substituting the two given conditions:

$$1 = -\frac{k}{2} + c$$

$$4 = -\frac{k}{8} + c$$

Subtracting the first equation from the second:

$$3 = \frac{3k}{8}$$

$$k = 8$$

$$c = 1 + \frac{8}{2} = 5$$

Hence

$$v^2 = 5 - \frac{16}{x^2}$$

(b) Since the particle moves in the direction of increasing x, use the positive square root from part (a). As $x > 0$, the velocity is

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v = \sqrt{5 - \frac{16}{x^2}} = \frac{\sqrt{5x^2 - 16}}{x}$$

Separating the variables and integrating:

$$\mathrm{d}t = \frac{x}{\sqrt{5x^2 - 16}}\,\mathrm{d}x$$

$$\int 1\,\mathrm{d}t = \frac{1}{10}\int 10x(5x^2 - 16)^{-1/2}\,\mathrm{d}x$$

$$t = \frac{1}{5}\sqrt{5x^2 - 16} + d$$

Let t_1 and t_2 be the times when $x = 2$ and $x = 4$, respectively. Substituting these values:

$$t_1 = \frac{1}{5}\sqrt{20 - 16} + d = \frac{2}{5} + d$$

$$t_2 = \frac{1}{5}\sqrt{80 - 16} + d = \frac{8}{5} + d$$

$$T = t_2 - t_1 = \frac{8}{5} - \frac{2}{5} = \frac{6}{5}$$

4. A particle P of mass $3\,\text{kg}$ moves along the positive x-axis. At time $t = 0$, it passes through the origin with speed $4\,\text{m}\,\text{s}^{-1}$.

While P is moving, the only force acting on it is a resistive force of magnitude

$$\frac{6}{v}\,\text{N}$$

where $v\,\text{m}\,\text{s}^{-1}$ is its speed. The force acts in the negative x-direction.

(a) Find an expression for v in terms of the displacement x of P from the origin. **[5]**

(b) Find the distance travelled by P while its speed decreases from $4\,\text{m}\,\text{s}^{-1}$ to $2\,\text{m}\,\text{s}^{-1}$. **[2]**

A particle Q, also of mass $3\,\text{kg}$, starts from the origin with speed $4\,\text{m}\,\text{s}^{-1}$. The only force acting on Q is a constant resistive force of magnitude $\frac{3}{2}\,\text{N}$.

(c) By comparing the resistive forces acting on P and Q, explain why P must come to rest at a time $t < 8\,\text{s}$ and at a displacement $x < 16\,\text{m}$. **[3]**

(d) Sketch the velocity–time graph for P from $t = 0$ until it first comes to rest. Your sketch should indicate the concavity of the graph, but no numerical values are required. **[1]**

(e) Determine the maximum displacement of P from the origin. **[2]**

Solution

(a) Take the direction of increasing x as positive. Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$3v\frac{\mathrm{d}v}{\mathrm{d}x} = -\frac{6}{v}$$

$$v^2\frac{\mathrm{d}v}{\mathrm{d}x} = -2$$

Separating the variables and integrating:

$$\int v^2\,\mathrm{d}v = -2\int 1\,\mathrm{d}x$$

$$\frac{v^3}{3} = -2x + c$$

When $x = 0$, $v = 4$, so $c = \frac{64}{3}$. Hence

$$v^3 = 64 - 6x$$

$$v = \sqrt[3]{64 - 6x}$$

(b) Substituting $v = 2$ into the result of part (a):

$$8 = 64 - 6x \quad \Longrightarrow \quad x = \frac{28}{3}\,\text{m}$$

Since P moves in the positive direction throughout, the distance travelled is $\frac{28}{3}\,\text{m}$.

(c) For Q, Newton's second law in the same positive direction gives

$$3a_Q = -\frac{3}{2} \quad \Longrightarrow \quad a_Q = -\frac{1}{2}\,\mathrm{m\,s^{-2}}$$

Using constant-acceleration formulae, its stopping time and displacement are

$$0 = 4 - \frac{1}{2}t \quad \Longrightarrow \quad t = 8\,\mathrm{s}$$

$$x_Q = 4(8) + \frac{1}{2}\left(-\frac{1}{2}\right)(8)^2 = 16\,\mathrm{m}$$

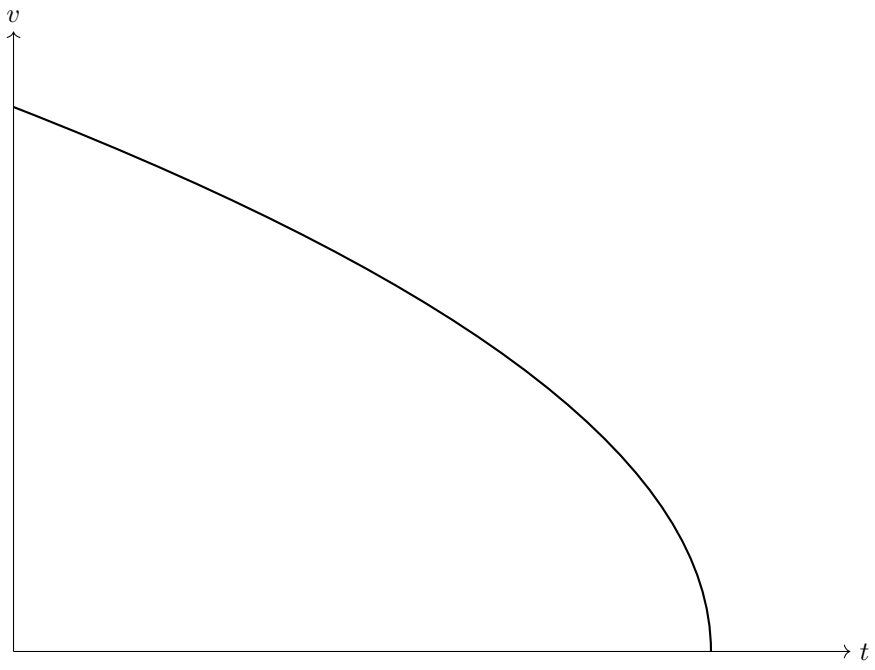
The initial resistive forces are equal. As P slows down, however, its resistive force increases:

$$0 < v < 4 \quad \Longrightarrow \quad \frac{6}{v} > \frac{3}{2}$$

Thus, at every speed below $4\,\mathrm{m\,s^{-1}}$, P has a greater deceleration than Q. Each further decrease in speed therefore takes less time and covers less distance for P, so it comes to rest with

$$t < 8\,\mathrm{s}, \qquad x < 16\,\mathrm{m}$$

(d) The sketch should look like:



As the speed falls, the resistive force increases, making the acceleration more negative. The graph is therefore decreasing and concave.

(e) The displacement increases while P is moving, so its maximum occurs as P comes to rest. Taking $v \to 0$ in part (a):

$$0 = 64 - 6x_{\max} \quad \Longrightarrow \quad x_{\max} = \frac{32}{3}\,\mathrm{m}$$

5. A particle B of mass 2 kg moves along the positive x-axis. When B passes through the origin, its speed is $u\,\mathrm{m\,s^{-1}}$.

The only horizontal force acting on B is a resistance whose magnitude is directly proportional to the cube of the speed of B. When the speed of B is $3\,\mathrm{m\,s^{-1}}$, the magnitude of the resistance is 54 N.

(a) Show that, while B is moving in the positive x-direction, its motion satisfies

$$\frac{1}{v^2}\frac{\mathrm{d}v}{\mathrm{d}x} = -1$$ **[3]**

(b) (i) Solve the differential equation in part (a) to obtain v in terms of x and u. **[4]**

(ii) By considering the behaviour of v as $x \to \infty$, state one respect in which this model of the motion is unrealistic. **[1]**

The particle subsequently reaches a point A with coordinate X, where its speed is $V\,\mathrm{m\,s^{-1}}$. The work done by the resistance as B moves from the origin to A is W J.

(c) (i) Using $W = \int F\,\mathrm{d}x$, where F is the signed force in the x-direction, find an expression for W in terms of X and u. **[3]**

(ii) Explain the physical significance of the sign of W. **[1]**

(iii) Rewrite your answer to part (c)(i) in terms of V and u, and hence show how W is related to the change in the kinetic energy of B. **[2]**

Solution

(a) Take the direction of increasing x as positive. Let the magnitude of the resistance be kv^3. The given force and speed give

$$54 = k(3^3) \quad \Longrightarrow \quad k = 2$$

Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$2v\frac{\mathrm{d}v}{\mathrm{d}x} = -kv^3 = -2v^3$$

While the particle is moving in the positive direction, $v > 0$, so dividing by $2v^3$ gives

$$\frac{1}{v^2}\frac{\mathrm{d}v}{\mathrm{d}x} = -1$$

(b) (i) Separating the variables and integrating:

$$\int v^{-2}\,\mathrm{d}v = -\int 1\,\mathrm{d}x$$

$$-\frac{1}{v} = -x + c$$

At the origin, $x = 0$ and $v = u$, so $c = -\frac{1}{u}$. Hence

$$-\frac{1}{v} = -x - \frac{1}{u}$$
$$\frac{1}{v} = x + \frac{1}{u} = \frac{1+ux}{u}$$
$$v = \frac{u}{1+ux}$$

(ii) As $x \to \infty$, $v \to 0$, but $v > 0$ for every finite x. Thus the model unrealistically predicts that the particle travels an unlimited distance without coming to rest.

(c) (i) The signed force is $F = -2v^3$. Using the speed from part (b)(i):

$$W = \int_0^X F \,\mathrm{d}x$$
$$= -2u^3 \int_0^X (1+ux)^{-3} \,\mathrm{d}x$$
$$= -2u^3 \left[\frac{(1+ux)^{-2}}{-2u}\right]_0^X$$
$$= u^2\left[(1+uX)^{-2} - 1\right]$$

(ii) $W < 0$: the resistance acts opposite to the displacement, so it does negative work and reduces the particle's kinetic energy.

(iii) From part (b)(i), $V = \frac{u}{1+uX}$, so the expression in part (c)(i) becomes

$$W = \frac{u^2}{(1+uX)^2} - u^2 = V^2 - u^2$$

Since the mass is 2 kg, the change in kinetic energy is

$$\frac{1}{2}(2)V^2 - \frac{1}{2}(2)u^2 = V^2 - u^2 = W$$

6. A small sphere P of mass $5\,\text{kg}$ is released into a liquid and moves vertically downwards. Take the downward direction as positive. In a model of the motion, the forces acting on P are:

- its weight;
- a constant upthrust of magnitude $14\,\text{N}$;
- a resistance to motion whose magnitude is proportional to the speed of P.

When the speed of P is $4\,\text{m s}^{-1}$, the resistance has magnitude $10\,\text{N}$. At time $t = 0$, the sphere is at a point O and is moving downwards with speed $u\,\text{m s}^{-1}$, where $u > 0$. At time t seconds, its downward velocity is $v\,\text{m s}^{-1}$. Take $g = 9.8\,\text{m s}^{-2}$.

(a) Show that, according to the model,

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 7 - \frac{1}{2}v$$ **[4]**

(b) (i) Solve the differential equation in part (a) to show that

$$v = 14 + (u - 14)e^{-t/2}$$ **[5]**

(ii) Describe briefly how the speed of P varies over time in each of the following cases:

- $0 < u < 14$;
- $u = 14$;
- $u > 14$.

In each case, state the limiting speed of P as $t \to \infty$. **[3]**

(c) Given that $u = 9$, find the exact downward displacement of P from O when $t = 6$. **[4]**

Solution

(a) Write the resistance as kv. Since $4k = 10$, $k = \frac{5}{2}$.

Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$5\frac{\mathrm{d}v}{\mathrm{d}t} = 5(9.8) - 14 - \frac{5}{2}v$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = 7 - \frac{1}{2}v$$

(b) (i) For $u \neq 14$, separate the variables in the equation from part (a):

$$\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{2}(v - 14)$$

$$\int \frac{1}{v - 14}\,\mathrm{d}v = -\frac{1}{2}\int 1\,\mathrm{d}t$$

$$\ln|v - 14| = -\frac{t}{2} + c$$

Using $v = u$ when $t = 0$ gives $c = \ln|u - 14|$. Hence

$$\ln|v - 14| - \ln|u - 14| = -\frac{t}{2}$$

$$|v - 14| = |u - 14|e^{-t/2}$$

The right-hand side is non-zero for finite t, so $v - 14$ cannot change sign. It therefore has the same sign as its initial value $u - 14$, giving

$$v - 14 = (u - 14)e^{-t/2}$$

If $u = 14$, the differential equation gives the constant solution $v = 14$. Thus, for every $u > 0$, the solution is

$$v = 14 + (u - 14)e^{-t/2}$$

(ii) Since $e^{-t/2}$ decreases from 1 towards 0:

- If $0 < u < 14$, the speed increases from u towards $14\,\mathrm{m\,s^{-1}}$.
- If $u = 14$, the speed remains constant at $14\,\mathrm{m\,s^{-1}}$.
- If $u > 14$, the speed decreases from u towards $14\,\mathrm{m\,s^{-1}}$.

In all three cases, the limiting speed is $14\,\mathrm{m\,s^{-1}}$.

(c) Let x be the downward displacement from O, so $x = 0$ when $t = 0$. Using $u = 9$ in part (b)(i):

$$\frac{\mathrm{d}x}{\mathrm{d}t} = v = 14 - 5e^{-t/2}$$

Integrating with respect to t:

$$x = 14t + 10e^{-t/2} + d$$

$$0 = 10 + d$$

$$d = -10$$

Therefore, at $t = 6$, the downward displacement is

$$x = 14(6) + 10e^{-3} - 10 = \left(74 + 10e^{-3}\right)\mathrm{m}$$

7. A vehicle of mass 900 kg moves along a straight horizontal road. Its engine provides a constant driving force of 576 N. When the vehicle has speed $v\,\mathrm{m\,s^{-1}}$, the resistance to its motion has magnitude (v^2+14v) N.

The time taken for the speed of the vehicle to increase from $3\,\mathrm{m\,s^{-1}}$ to $13\,\mathrm{m\,s^{-1}}$ is T seconds.

(a) Show that

$$T = 18\ln\left(\frac{27}{7}\right)$$ **[6]**

When $t = 0$, the speed of the vehicle is $8\,\mathrm{m\,s^{-1}}$.

(b) Show that

$$\frac{v+32}{18-v} = pe^{qt}$$

where the rational numbers p and q are to be found. **[2]**

(c) Hence explain why $v < 18$ for all $t \geq 0$. **[1]**

Solution

(a) Take the direction of motion as positive. Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$\begin{aligned} 900\frac{\mathrm{d}v}{\mathrm{d}t} &= 576 - (v^2 + 14v) \\ &= (18-v)(v+32) \end{aligned}$$

Separating the variables:

$$\frac{900}{(18-v)(v+32)}\,\mathrm{d}v = \mathrm{d}t$$

For the partial fractions, write

$$\frac{1}{(18-v)(v+32)} = \frac{A}{18-v} + \frac{B}{v+32}$$

Multiplying by $(18-v)(v+32)$ and comparing coefficients gives

$$\begin{aligned} 1 &= A(v+32) + B(18-v) \\ &= (A-B)v + 32A + 18B \end{aligned}$$

$$A - B = 0, \qquad 32A + 18B = 1$$

$$A = B = \frac{1}{50}$$

Both $18-v$ and $v+32$ are positive while $0 \leq v < 18$, which part (c) shows holds throughout the motion. Integrating both sides:

$$18\int\left(\frac{1}{18-v} + \frac{1}{v+32}\right)\mathrm{d}v = \int 1\,\mathrm{d}t$$

$$18(-\ln(18-v) + \ln(v+32)) = t + c$$

Combining the logarithms:

$$18\ln\left(\frac{v+32}{18-v}\right) = t + c$$

Let t_3 and t_{13} be the times at which the speeds are 3 and $13\,\mathrm{m\,s^{-1}}$, respectively. Substituting these speeds:

$$t_3 + c = 18\ln\left(\frac{35}{15}\right) = 18\ln\left(\frac{7}{3}\right)$$

$$t_{13} + c = 18\ln\left(\frac{45}{5}\right) = 18\ln 9$$

Subtracting eliminates c:

$$T = t_{13} - t_3 = 18\left(\ln 9 - \ln\left(\frac{7}{3}\right)\right)$$

$$= 18\ln\left(\frac{27}{7}\right)$$

(b) Using the integrated relation from part (a), the condition $v = 8$ when $t = 0$ gives

$$c = 18\ln\left(\frac{8+32}{18-8}\right) = 18\ln 4$$

Thus, rearranging and exponentiating:

$$\ln\left(\frac{v+32}{18-v}\right) = \frac{t}{18} + \ln 4$$

$$\frac{v+32}{18-v} = 4e^{t/18}$$

This is the required form pe^{qt}, with

$$p = 4, \qquad q = \frac{1}{18}$$

(c) From part (b), $v + 32 = 4e^{t/18}(18 - v)$. Since $v + 32 = 50 - (18 - v)$, this rearranges to

$$18 - v = \frac{50}{1 + 4e^{t/18}} > 0$$

The denominator is positive for every $t \geq 0$, so $v < 18$ for all $t \geq 0$.

8. A particle P of mass $2\,\text{kg}$ moves along the positive x-axis in the direction of increasing x. At time t seconds, its displacement from the origin is x metres and its speed is $v\,\text{m s}^{-1}$.

While P is moving, the resultant force acting on it is directed towards the origin and has magnitude $kv(v+2)\,\text{N}$, where k is a positive constant and v denotes the numerical value of the speed in metres per second.

When $x = 1$, the speed of P is $7\,\text{m s}^{-1}$, and when $x = 3$, its speed is $1\,\text{m s}^{-1}$.

(a) Find k and show that

$$v = 9 \times 3^{(1-x)/2} - 2$$ **[6]**

The time taken for the speed of P to decrease from $7\,\text{m s}^{-1}$ to $1\,\text{m s}^{-1}$ is T seconds.

(b) Show that

$$T = \frac{\ln(7/3)}{\ln 3}$$ **[5]**

Solution

(a) Take the direction of increasing x as positive. The resultant force acts in the negative direction. Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$2v\frac{\mathrm{d}v}{\mathrm{d}x} = -kv(v+2)$$

Since $v > 0$ while the particle is moving, $v + 2 > 0$. Dividing by v and separating the variables:

$$\frac{1}{v+2}\,\mathrm{d}v = -\frac{k}{2}\,\mathrm{d}x$$

$$\int \frac{1}{v+2}\,\mathrm{d}v = -\frac{k}{2}\int 1\,\mathrm{d}x$$

$$\ln(v+2) = -\frac{kx}{2} + c$$

Substituting the two given conditions:

$$\ln 9 = -\frac{k}{2} + c$$

$$\ln 3 = -\frac{3k}{2} + c$$

Subtracting the first equation from the second:

$$\ln 3 - \ln 9 = -k$$

$$k = \ln 3$$

Subtracting the relation at $x = 1$, $v = 7$ from the general relation now gives

$$\ln(v+2) - \ln 9 = -\frac{k}{2}(x-1)$$

$$\ln\left(\frac{v+2}{9}\right) = \frac{1-x}{2}\ln 3$$

$$\frac{v+2}{9} = 3^{(1-x)/2}$$

$$v = 9 \times 3^{(1-x)/2} - 2$$

(b) Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$2\frac{\mathrm{d}v}{\mathrm{d}t} = -kv(v+2)$$

Both v and $v+2$ are positive. Separating the variables, and using partial fractions:

$$\frac{2}{v(v+2)}\,\mathrm{d}v = -k\,\mathrm{d}t$$

$$\frac{2}{v(v+2)} = \frac{1}{v} - \frac{1}{v+2}$$

$$\int\left(\frac{1}{v} - \frac{1}{v+2}\right)\mathrm{d}v = -k\int 1\,\mathrm{d}t$$

$$\ln v - \ln(v+2) = -kt + d$$

Combining the logarithms:

$$\ln\left(\frac{v}{v+2}\right) = -kt + d$$

Let t_0 be the time when $v = 7$. Then $v = 1$ at time $t_0 + T$, giving

$$\ln\left(\frac{7}{9}\right) = -kt_0 + d$$

$$\ln\left(\frac{1}{3}\right) = -k(t_0+T) + d$$

Subtracting eliminates d and t_0. Using $k = \ln 3$ from part (a):

$$\ln\left(\frac{1}{3}\right) - \ln\left(\frac{7}{9}\right) = -kT$$

$$\ln\left(\frac{3}{7}\right) = -T\ln 3$$

$$T = \frac{\ln(7/3)}{\ln 3}$$

9. A particle P of mass 4 kg is released from rest and moves vertically downwards through a fluid. While P is moving with speed $v\,\mathrm{m\,s^{-1}}$, it experiences a resistance of magnitude $6v\,\mathrm{N}$ acting vertically upwards. After P has descended a distance $x\,\mathrm{m}$, its speed is $v\,\mathrm{m\,s^{-1}}$. Take $g = 9.8\,\mathrm{m\,s^{-2}}$.

(a) Show that

$$\left(\frac{10v}{98-15v}\right)\frac{\mathrm{d}v}{\mathrm{d}x} = 1 \qquad \textbf{[2]}$$

(b) Verify that x and v satisfy

$$45x = 196\ln\left(\frac{98}{98-15v}\right) - 30v \qquad \textbf{[5]}$$

(c) Determine the work done against the resistance from the instant P is released until its acceleration has magnitude $2.45\,\mathrm{m\,s^{-2}}$. Give your answer to 3 significant figures. **[5]**

Solution

(a) Take vertically downwards as positive. Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$4v\frac{\mathrm{d}v}{\mathrm{d}x} = 4(9.8) - 6v$$

$$10v\frac{\mathrm{d}v}{\mathrm{d}x} = 98 - 15v$$

$$\left(\frac{10v}{98-15v}\right)\frac{\mathrm{d}v}{\mathrm{d}x} = 1$$

(b) Separating the variables in the equation from part (a):

$$\int \frac{10v}{98-15v}\,\mathrm{d}v = \int 1\,\mathrm{d}x$$

The terminal speed is $\frac{98}{15}\,\mathrm{m\,s^{-1}}$. Since the particle accelerates towards this speed from rest, $98-15v > 0$. To integrate, write

$$\frac{10v}{98-15v} = -\frac{2}{3} + \frac{196}{3(98-15v)}$$

Hence

$$x = -\frac{2v}{3} - \frac{196}{45}\ln(98-15v) + c$$

Initially, $x = 0$ and $v = 0$, so

$$0 = -\frac{196}{45}\ln 98 + c \quad\Longrightarrow\quad c = \frac{196}{45}\ln 98$$

Substituting for c and combining the logarithms:

$$45x = -30v - 196\ln(98-15v) + 196\ln 98$$

$$= 196\ln\left(\frac{98}{98-15v}\right) - 30v$$

(c) The acceleration is downwards throughout the motion. Newton's second law gives

$$a = 9.8 - 1.5v$$

Setting its magnitude to $2.45\,\mathrm{m\,s^{-2}}$:

$$2.45 = 9.8 - 1.5v \quad \Longrightarrow \quad v = 4.9\,\mathrm{m\,s^{-1}}$$

Using part (b), the distance descended is

$$45x = 196\ln\left(\frac{98}{98 - 15(4.9)}\right) - 30(4.9)$$

$$x = \frac{196\ln 4 - 147}{45}\,\mathrm{m}$$

Let W be the positive work done against resistance. The work done by resistance is $-W$. The work–energy principle, with work positive when a force acts in the direction of motion, is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

Take the final position as the zero level for gravitational potential energy. Substituting:

$$\begin{aligned} 0 + 4gx - W &= \frac{1}{2}(4)v^2 + 0 \\ W &= 39.2x - 2v^2 \\ &= 39.2\left(\frac{196\ln 4 - 147}{45}\right) - 2(4.9)^2 \\ &\approx 60.6\,\mathrm{J} \qquad (3\text{ s.f.}) \end{aligned}$$

10. A particle P, of mass m, moves on a rough straight track inclined at an angle α to the horizontal. A fixed point O lies on the track, and x denotes the distance of P from O, measured up the track.

In addition to its weight, P is acted on by a force directed down the track and having magnitude

$$\frac{kmg}{x+a}$$

where a and k are positive constants with dimensions of length. The coefficient of friction between P and the track is μ.

The particle is projected up the track and first comes to rest at the point where $x = 2a$.

(a) Show that, while P is moving up the track with speed v,

$$\frac{\mathrm{d}}{\mathrm{d}x}(v^2) + \frac{2gk}{x+a} + 2g(\sin\alpha + \mu\cos\alpha) = 0 \qquad \textbf{[4]}$$

(b) Verify that

$$v^2 = 2gk\ln\left(\frac{3a}{x+a}\right) + 2g(\sin\alpha + \mu\cos\alpha)(2a - x) \qquad \textbf{[4]}$$

(c) Find, in terms of a, k and α, the range of values of μ for which P remains at rest when $x = 2a$. **[2]**

Solution

(a) Take up the track as positive. Let the normal reaction be R. Resolving perpendicular to the track gives

$$R = mg\cos\alpha$$

While P moves up the track, friction acts down the track with magnitude μR. Using Newton's second law, with acceleration $v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$mv\frac{\mathrm{d}v}{\mathrm{d}x} = -mg\sin\alpha - \mu R - \frac{kmg}{x+a}$$

$$2v\frac{\mathrm{d}v}{\mathrm{d}x} = -2g(\sin\alpha + \mu\cos\alpha) - \frac{2gk}{x+a}$$

Since $\frac{\mathrm{d}}{\mathrm{d}x}(v^2) = 2v\frac{\mathrm{d}v}{\mathrm{d}x}$, this becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}(v^2) + \frac{2gk}{x+a} + 2g(\sin\alpha + \mu\cos\alpha) = 0$$

(b) Since x is a distance and $a > 0$, $x + a > 0$. Integrating the equation in part (a) with respect to x:

$$v^2 = -2gk\ln(x+a) - 2g(\sin\alpha + \mu\cos\alpha)x + c$$

Substituting $v = 0$ when $x = 2a$ gives

$$0 = -2gk\ln(3a) - 4ga(\sin\alpha + \mu\cos\alpha) + c$$

$$c = 2gk\ln(3a) + 4ga(\sin\alpha + \mu\cos\alpha)$$

Hence

$$v^2 = 2gk\big(\ln(3a) - \ln(x+a)\big) + 2g(\sin\alpha + \mu\cos\alpha)(2a - x)$$

$$= 2gk\ln\left(\frac{3a}{x+a}\right) + 2g(\sin\alpha + \mu\cos\alpha)(2a - x)$$

(c) At $x = 2a$, let the frictional force acting up the track be F. For equilibrium, Newton's second law gives

$$F - mg\sin\alpha - \frac{kmg}{3a} = 0$$

The required friction must not exceed its limiting value $\mu R = \mu mg\cos\alpha$, so

$$mg\sin\alpha + \frac{kmg}{3a} \leq \mu mg\cos\alpha$$

$$\mu \geq \tan\alpha + \frac{k}{3a\cos\alpha}$$

11. A small instrument package of mass $m\,\text{kg}$ enters a deep liquid moving vertically downwards at $7\,\text{m s}^{-1}$. While the package is moving downwards, it experiences a resistive force of magnitude $\frac{m}{20}v^2\,\text{N}$, where $v\,\text{m s}^{-1}$ is its speed.

At time $t\,\text{s}$ after entering the liquid, the package has descended a distance $x\,\text{m}$.

(a) Find an expression for v in terms of t. **[5]**

(b) Find an expression for x in terms of t. **[4]**

(c) Find the exact distance descended by the package when its speed first reaches $\frac{21}{2}\,\text{m s}^{-1}$. **[4]**

Solution

(a) Take vertically downwards as positive. Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - \frac{m}{20}v^2$$

$$20\frac{\mathrm{d}v}{\mathrm{d}t} = 196 - v^2$$

The terminal speed is $14\,\text{m s}^{-1}$. Since the initial speed is $7\,\text{m s}^{-1}$, the speed increases towards $14\,\text{m s}^{-1}$ and remains below it, so $14 + v$ and $14 - v$ are both positive.

Separating the variables and using partial fractions:

$$\int \frac{1}{196 - v^2}\,\mathrm{d}v = \frac{1}{20}\int 1\,\mathrm{d}t$$

$$\frac{1}{28}\int\left(\frac{1}{14 - v} + \frac{1}{14 + v}\right)\mathrm{d}v = \frac{t}{20} + c$$

$$\frac{1}{28}(\ln(14 + v) - \ln(14 - v)) = \frac{t}{20} + c$$

Using $v = 7$ when $t = 0$:

$$c = \frac{1}{28}\ln\left(\frac{21}{7}\right) = \frac{\ln 3}{28}$$

Hence

$$\ln\left(\frac{14 + v}{14 - v}\right) = \frac{7t}{5} + \ln 3$$

$$\frac{14 + v}{14 - v} = 3e^{7t/5}$$

$$14 + v = 42e^{7t/5} - 3ve^{7t/5}$$

$$v\left(1 + 3e^{7t/5}\right) = 14\left(3e^{7t/5} - 1\right)$$

$$v = \frac{14\left(3e^{7t/5} - 1\right)}{3e^{7t/5} + 1}$$

(b) Using $v = \frac{\mathrm{d}x}{\mathrm{d}t}$, rewrite the result from part (a), whose denominator $3e^{7t/5} + 1$ is positive, to recognise a logarithmic integral:

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}t} &= -14 + \frac{84e^{7t/5}}{3e^{7t/5}+1} \\ &= -14 + 20\frac{\frac{21}{5}e^{7t/5}}{3e^{7t/5}+1} \\ x &= -14t + 20\ln\left(3e^{7t/5}+1\right) + d\end{aligned}$$

Since $x = 0$ when $t = 0$, $d = -20\ln 4$. Therefore

$$x = 20\ln\left(\frac{3e^{7t/5}+1}{4}\right) - 14t$$

(c) Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{196 - v^2}{20}$$

Separating the variables and integrating, with $196 - v^2 > 0$:

$$\begin{aligned}\int 1\,\mathrm{d}x &= \int \frac{20v}{196-v^2}\,\mathrm{d}v \\ x &= -10\ln(196 - v^2) + h\end{aligned}$$

At entry, $x = 0$ and $v = 7$, so $h = 10\ln 147$. Thus, when $v = \frac{21}{2}$, the distance descended is

$$x = 10\ln\left(\frac{147}{196 - \left(\frac{21}{2}\right)^2}\right) = 10\ln\left(\frac{12}{7}\right)\text{ m}$$

Final answers with $g = 10$: (a) $v = 10\sqrt{2}\,\frac{(10\sqrt{2}+7)e^{\sqrt{2}t} - (10\sqrt{2}-7)}{(10\sqrt{2}+7)e^{\sqrt{2}t} + (10\sqrt{2}-7)}$;
(b) $x = 20\ln\left(\frac{(10\sqrt{2}+7)e^{\sqrt{2}t} + 10\sqrt{2} - 7}{20\sqrt{2}}\right) - 10\sqrt{2}\,t$; (c) $10\ln\left(\frac{604}{359}\right)$ m

12. A package of mass m kg is released from rest and falls vertically. When its downward speed is $v\,\mathrm{m\,s^{-1}}$, it experiences a resistive force of magnitude mkv^2 N, where k is a positive constant. Take the downward direction as positive.

(a) Show that, at time t s after release,

$$v = \sqrt{\frac{g}{k}}\left(\frac{e^{2\sqrt{gk}\,t} - 1}{e^{2\sqrt{gk}\,t} + 1}\right) \qquad \textbf{[5]}$$

It is given that $k = 0.0125\,\mathrm{m^{-1}}$.

(b) Find the distance fallen by the package during the first 2 seconds after its release. **[4]**

Solution

(a) Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = mg - mkv^2$$

$$\frac{\mathrm{d}v}{\mathrm{d}t} = g - kv^2$$

Let $V = \sqrt{g/k}$ be the terminal speed. The package accelerates from rest towards this speed, so $0 \le v < V$, and $V + v$ and $V - v$ are both positive.

Separating the variables and using partial fractions:

$$\int \frac{1}{V^2 - v^2}\,\mathrm{d}v = \int k\,\mathrm{d}t$$

$$\frac{1}{2V}\int\left(\frac{1}{V+v} + \frac{1}{V-v}\right)\mathrm{d}v = \int k\,\mathrm{d}t$$

$$\frac{1}{2V}\ln\left(\frac{V+v}{V-v}\right) = kt + c$$

At release, $t = 0$ and $v = 0$, giving $c = 0$. Hence

$$\ln\left(\frac{V+v}{V-v}\right) = 2kVt$$

$$V + v = e^{2kVt}(V - v)$$

$$v\left(1 + e^{2kVt}\right) = V\left(e^{2kVt} - 1\right)$$

$$v = \sqrt{\frac{g}{k}}\left(\frac{e^{2\sqrt{gk}\,t} - 1}{e^{2\sqrt{gk}\,t} + 1}\right)$$

(b) Let s be the distance fallen after t seconds. Using $g = 9.8$ and $k = 0.0125$ in part (a):

$$\frac{\mathrm{d}s}{\mathrm{d}t} = v = 28\left(\frac{e^{0.7t} - 1}{e^{0.7t} + 1}\right)$$

$$= \frac{56e^{0.7t}}{e^{0.7t} + 1} - 28$$

The denominator $e^{0.7t}+1$ is positive. Integrating by recognising the derivative of the denominator:

$$\begin{aligned} s &= \int \left(\frac{56e^{0.7t}}{e^{0.7t}+1} - 28 \right) \mathrm{d}t \\ &= 80\ln\left(e^{0.7t}+1\right) - 28t + d \end{aligned}$$

Using $s = 0$ when $t = 0$ gives $d = -80\ln 2$, so

$$s = 80\ln\left(\frac{e^{0.7t}+1}{2}\right) - 28t$$

The distance fallen in the first 2 seconds is therefore

$$s(2) = \left(80\ln\left(\frac{e^{1.4}+1}{2}\right) - 56\right)\text{ m} \approx 18.2\text{ m}$$

Final answers with $g = 10$: (b) $\left(80\ln\left(\frac{e^{\sqrt{2}}+1}{2}\right) - 40\sqrt{2}\right)\text{ m} \approx 18.5\text{ m}$

13. A particle P of mass 16 kg moves in a straight line on a smooth horizontal plane.

At time $t = 0$, the particle passes through a fixed point O with velocity $\frac{5\sqrt{3}}{4}\,\mathrm{m\,s^{-1}}$.

From this instant, a horizontal force opposing the motion acts on P. The magnitude of the force is proportional to $\sqrt{25 - 4v^2}$, where $v\,\mathrm{m\,s^{-1}}$ is the velocity of P.

When $v = 2$, the magnitude of the force is 6 N. The force ceases to act when the velocity has decreased to $\frac{5}{4}\,\mathrm{m\,s^{-1}}$.

(a) Show that, while the force is acting, the motion of P is modelled by the differential equation

$$\frac{1}{\sqrt{25 - 4v^2}}\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{8} \qquad \textbf{[3]}$$

(b) Hence find v in terms of t, showing that your expression is valid for

$$0 \leq t \leq \frac{2\pi}{3} \qquad \textbf{[5]}$$

(c) Determine the total distance travelled by P during the interval $0 \leq t \leq 8$. Give your answer in an exact form. **[6]**

Solution

(a) Take the direction of motion as positive. Let the magnitude of the opposing force be $F = k\sqrt{25 - 4v^2}$, where $k > 0$.

Using the given force when $v = 2$:

$$6 = k\sqrt{25 - 4(2)^2}$$

$$6 = 3k$$

$$k = 2$$

Using Newton's second law, with acceleration $a = \frac{\mathrm{d}v}{\mathrm{d}t}$:

$$16\frac{\mathrm{d}v}{\mathrm{d}t} = -2\sqrt{25 - 4v^2}$$

$$\frac{1}{\sqrt{25 - 4v^2}}\frac{\mathrm{d}v}{\mathrm{d}t} = -\frac{1}{8}$$

(b) The standard result is

$$\int \frac{1}{\sqrt{b^2 - v^2}}\,\mathrm{d}v = \arcsin\left(\frac{v}{b}\right) + c \qquad (|v| < b)$$

Separating the variables, factoring out the numerical scale and using it with $b = \frac{5}{2}$:

$$\frac{\mathrm{d}v}{\sqrt{25 - 4v^2}} = -\frac{1}{8}\,\mathrm{d}t$$

$$\frac{1}{2}\int \frac{1}{\sqrt{\left(\frac{5}{2}\right)^2 - v^2}}\,\mathrm{d}v = -\frac{1}{8}\int 1\,\mathrm{d}t$$

$$\frac{1}{2}\arcsin\left(\frac{2v}{5}\right) = -\frac{t}{8} + c$$

Substituting $t = 0$ and $v = \frac{5\sqrt{3}}{4}$:

$$c = \frac{1}{2}\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6}$$

Hence

$$\arcsin\left(\frac{2v}{5}\right) = \frac{\pi}{3} - \frac{t}{4}$$

The force ceases when $v = \frac{5}{4}$, so

$$\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{3} - \frac{t}{4}$$

$$\frac{\pi}{6} = \frac{\pi}{3} - \frac{t}{4}$$

$$t = \frac{2\pi}{3}$$

Taking sine gives the required velocity while the force acts:

$$v = \frac{5}{2}\sin\left(\frac{\pi}{3} - \frac{t}{4}\right), \qquad 0 \le t \le \frac{2\pi}{3}$$

(c) Let x be the displacement from O in the positive direction. Using $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ and integrating while the force acts:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{5}{2}\sin\left(\frac{\pi}{3} - \frac{t}{4}\right)$$

$$x = 10\cos\left(\frac{\pi}{3} - \frac{t}{4}\right) + d$$

Substituting $x = 0$ at $t = 0$:

$$0 = 10\cos\frac{\pi}{3} + d \quad \Longrightarrow \quad d = -5$$

The velocity remains positive, so the distance travelled while the force acts is

$$x\left(\frac{2\pi}{3}\right) = 10\cos\frac{\pi}{6} - 5 = 5\sqrt{3} - 5$$

After the force ceases, there is no horizontal force, so the particle continues at the constant velocity $\frac{5}{4}\,\mathrm{m\,s^{-1}}$. Therefore

$$\begin{aligned}\text{Total distance} &= (5\sqrt{3}-5)+\frac{5}{4}\left(8-\frac{2\pi}{3}\right)\\ &= \left(5\sqrt{3}+5-\frac{5\pi}{6}\right)\mathrm{m}\end{aligned}$$

14. A vehicle of mass m kg moves along a horizontal straight road. Its engine works at a constant rate of $2kmg$ W, where k is a positive constant with units of speed. When the vehicle has speed $v\,\mathrm{m\,s^{-1}}$, it experiences a resistance of magnitude $\frac{mgv}{k}$ N.

Initially, the vehicle is at rest at $x = 0$, where x m is the distance travelled.

(a) Show that

$$v^2\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{g}{k}\left(2k^2 - v^2\right) \qquad \textbf{[3]}$$

(b) Verify that

$$gx = -kv + \frac{k^2}{\sqrt{2}}\ln\left(\frac{\sqrt{2}k + v}{\sqrt{2}k - v}\right) \qquad \textbf{[5]}$$

(c) By using the work–energy principle, show that the time taken for the vehicle to accelerate from rest to a speed V, where $0 < V < \sqrt{2}k$, is

$$\frac{k}{2g}\ln\left(\frac{2k^2}{2k^2 - V^2}\right) \qquad \textbf{[8]}$$

Solution

(a) Take the direction of motion as positive. Using $P = Fv$, the driving force is

$$F = \frac{2kmg}{v}$$

Using Newton's second law, with acceleration $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$:

$$mv\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{2kmg}{v} - \frac{mgv}{k}$$

$$v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{g}{kv}\left(2k^2 - v^2\right)$$

$$v^2\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{g}{k}\left(2k^2 - v^2\right)$$

(b) Separating the variables in part (a):

$$g\,\mathrm{d}x = k\frac{v^2}{2k^2 - v^2}\,\mathrm{d}v$$

Use algebraic division followed by partial fractions:

$$\frac{v^2}{2k^2 - v^2} = -1 + \frac{2k^2}{2k^2 - v^2}$$

$$= -1 + \frac{k}{\sqrt{2}}\left(\frac{1}{\sqrt{2}k - v} + \frac{1}{\sqrt{2}k + v}\right)$$

The resultant force is zero at the terminal speed $\sqrt{2}k$. During acceleration from rest, $0 \le v < \sqrt{2}k$, so $\sqrt{2}k + v$ and $\sqrt{2}k - v$ are both positive. Integrating gives

$$\begin{aligned} gx &= k \int \left[-1 + \frac{k}{\sqrt{2}} \left(\frac{1}{\sqrt{2}k - v} + \frac{1}{\sqrt{2}k + v} \right) \right] \mathrm{d}v \\ &= -kv + \frac{k^2}{\sqrt{2}} \Big(\ln(\sqrt{2}k + v) - \ln(\sqrt{2}k - v) \Big) + c \end{aligned}$$

When $x = 0$, $v = 0$, giving

$$0 = \frac{k^2}{\sqrt{2}} \Big(\ln(\sqrt{2}k) - \ln(\sqrt{2}k) \Big) + c \quad \Longrightarrow \quad c = 0$$

Combining the logarithms:

$$gx = -kv + \frac{k^2}{\sqrt{2}} \ln\left(\frac{\sqrt{2}k + v}{\sqrt{2}k - v} \right)$$

(c) Let $W(v)$ be the work done by resistance from rest until the speed is v. This work is negative because resistance opposes the motion. From part (a):

$$\mathrm{d}x = \frac{kv^2}{g(2k^2 - v^2)} \, \mathrm{d}v$$

Hence

$$\begin{aligned} \mathrm{d}W &= -\frac{mgv}{k} \, \mathrm{d}x \\ \frac{\mathrm{d}W}{\mathrm{d}v} &= -\frac{mv^3}{2k^2 - v^2} = mv - \frac{2mk^2 v}{2k^2 - v^2} \end{aligned}$$

Since $0 \le v \le V < \sqrt{2}k$, $2k^2 - v^2 > 0$. Integrating by recognising the derivative of $2k^2 - v^2$:

$$W(v) = \frac{1}{2}mv^2 + mk^2 \ln(2k^2 - v^2) + d$$

Using $W(0) = 0$ gives $d = -mk^2 \ln(2k^2)$, so

$$W(V) = \frac{1}{2}mV^2 - mk^2 \ln\left(\frac{2k^2}{2k^2 - V^2} \right)$$

The work–energy principle, with work positive when a force acts in the direction of motion, is

$$KE_0 + GPE_0 + \sum W_{\text{other}} = KE_1 + GPE_1$$

Let the required time be t. The engine does work $Pt = 2kmg\,t$. There is no change in gravitational potential energy. Substituting:

$$\begin{aligned} 0 + 2kmg\,t + W(V) &= \frac{1}{2}mV^2 \\ 2kmg\,t + \frac{1}{2}mV^2 - mk^2 \ln\left(\frac{2k^2}{2k^2 - V^2} \right) &= \frac{1}{2}mV^2 \\ 2kmg\,t &= mk^2 \ln\left(\frac{2k^2}{2k^2 - V^2} \right) \\ t &= \frac{k}{2g} \ln\left(\frac{2k^2}{2k^2 - V^2} \right) \end{aligned}$$